SecureDT - Review 0 Guide

Verification checklist for reviewers

Deliverable status

SecureDT is a complete, runnable Flask prototype. The following defines what a review team should see and how to verify each claim.

A. Run the test suite

```
python tests/run_all.py
```

Expect: 44+ tests passing covering datasets, digital twin, risk calibration, GA convergence, schedulers, attacks, storage, AI and API/security defence.

B. Verify risk calibration

```
python -c "from engine.digital_twin import DigitalTwin;
print(DigitalTwin().apply_attack('MEC-2','DDoS',0.8)['after'])"
```

Expect security_risk > 0.8 and status CRITICAL.

C. Verify GA convergence

```
python engine/scheduler.py
```

Watch best-fitness decrease (improve) over generations.

D. Verify the live dashboard

- Start python app.py and open /
- Generate tasks, run GA, launch an attack on MEC-2, mitigate, then observe the security timeline update
- Open /application-security and /system-health to confirm headers, CSRF and services are reported

E. Known limitations

- Single-role / session-ready architecture (not multi-user auth)
- AI model is offline and trained on synthetic data
- All attacks are simulated; no external systems are contacted